

A Multi-Source Retail Intelligence Platform Using Data Warehousing, Customer Segmentation, Market Basket Analysis, and Holiday-Aware Analytics

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Abstract—This paper presents a multi-source Retail Intelligence Platform applied to the Dunnhumby Complete Journey dataset, comprising 1,469,307 transactions from 2,500 households across 302 product categories. The platform integrates a U.S. Federal Holiday dimension and implements a three-layer Azure PostgreSQL data warehouse supporting Market Basket Analysis (MBA) and customer segmentation. We address three specific problems: (P1) the absence of an empirical algorithm benchmark at grocery scale; (P2) the systematic failure of K-Means to detect high-value customer outliers; and (P3) reliance on the unverified assumption that holiday periods drive higher retail sales. Three MBA algorithms—Apriori, FP-Growth, and ECLAT—and three segmentation algorithms—K-Means, Hierarchical Clustering, and DBSCAN—are implemented and compared. Experimental results establish that all three MBA algorithms converge on identical rule sets (104,567 frequent itemsets, 98,853 rules, mean lift 4.272, maximum lift 25.49), while ECLAT achieves a $13.7\times$ runtime reduction over Apriori (26.58 s vs. 364.22 s), explained by the dataset’s 2.3% basket sparsity. DBSCAN achieves a silhouette score of 0.758 versus K-Means’s 0.656 and uniquely identifies 55 VIP customers each generating \$10,304/year—a \$566,720 annual misclassification that K-Means systematically produces. Integration of a Federal Holiday dimension reveals a -7.95% holiday sales differential, empirically refuting the standard retail uplift assumption. Results are operationalized through a five-page Power BI dashboard and an association-rule-backed Flask recommendation prototype.

Index Terms—Market Basket Analysis, Apriori, FP-Growth, ECLAT, Customer Segmentation, DBSCAN, RFM, Data Warehouse, Azure PostgreSQL, Power BI, Retail Analytics, Dunnhumby, Holiday Analytics

I. INTRODUCTION

Retail organizations accumulate large volumes of transactional, product, household, coupon, campaign, and promotion data. Despite this abundance, raw retail data is typically fragmented across operational files, contains quality defects, and is not directly suitable for business intelligence or advanced analytics. Transforming such data into actionable insights requires an integrated pipeline spanning data engineering, warehousing, behavioral mining, and reporting.

This paper presents a Retail Intelligence Platform that addresses this challenge by integrating the Dunnhumby Complete Journey dataset [8] with a U.S. Federal Holidays dataset into a production-grade Azure PostgreSQL data warehouse.

Beyond descriptive analytics, the platform applies MBA and customer segmentation to surface findings that would not be visible to a retailer operating from raw transaction data alone.

Three research problems motivate this work. First, grocery retail practitioners lack empirical evidence for MBA algorithm selection at scale, defaulting to Apriori despite the existence of more efficient alternatives. Second, K-Means—the dominant retail segmentation method—has a structural limitation in outlier detection whose monetary consequences have not been quantified in the grocery context. Third, promotional calendars routinely assume that federal holidays drive increased purchasing, an assumption that has not been empirically tested on this dataset. The platform’s experimental results produce specific, falsifiable findings for each of these three problems.

The remainder of this paper is organized as follows. Section II reviews related work. Section III identifies the research gap. Section IV describes the system architecture. Section V covers data sources and preprocessing. Section VI presents the data warehouse design. Section VII describes the analytical methodology. Section VIII presents experimental results. Section IX discusses key findings and business implications. Section XII concludes with future directions.

II. RELATED WORK

A. Data Warehousing and Dimensional Modeling

Kimball and Ross established dimensional modeling principles based on fact and dimension tables, enabling efficient analytical querying and business reporting [13]. Their galaxy and star schema patterns are widely adopted for retail decision-support systems. This work implements a galaxy/star-inspired schema under Azure PostgreSQL, extending the classical model with a holiday dimension not present in the source dataset.

B. Association Rule Mining

Agrawal et al. [1] introduced association rule mining for discovering relationships between itemsets in large databases. Apriori generates candidate itemsets level-by-level through repeated database scans, becoming expensive as candidate counts grow. Han et al. [2] proposed FP-Growth, which

compresses transactions into an FP-tree enabling single-pass pattern extraction without candidate generation. Zaki [3] introduced ECLAT, a depth-first algorithm using vertical transaction ID set (tidset) intersections. Borgelt [4] demonstrated FP-Growth’s memory advantages on dense datasets, while Fournier-Viger et al. [5] extended comparisons to modern hardware. Prior benchmarks were conducted on synthetic or small UCI datasets; this work provides the first three-way comparison on the full Dunnhumby dataset at 155K+ baskets with simultaneous runtime and memory profiling.

C. Customer Segmentation

Hughes [10] formalized RFM (Recency, Frequency, Monetary) analysis as a segmentation framework. K-Means-based RFM segmentation has since become the retail industry standard [11], [12], with typical silhouette scores of 0.45–0.65 reported on grocery datasets. Ester et al. [9] introduced DBSCAN, which identifies clusters through density thresholds and treats low-density points as outliers. While DBSCAN has been applied in geographic and network contexts, its application to RFM retail segmentation for automatic VIP detection has not been widely studied. This work establishes DBSCAN as a superior alternative to K-Means and quantifies the revenue impact of K-Means’s structural failure mode.

D. Retail Market Basket Analysis on Real Datasets

Brijs et al. [6] applied MBA to Belgian retail data, reporting lift values of 2–8 for grocery co-purchase patterns. Liao et al. [7] applied FP-Growth to a Taiwanese supermarket chain with lift values of 3–12. The Dunnhumby Complete Journey dataset was published in 2014 [8] as a retail analytics benchmark, but no published study has simultaneously applied all three major MBA algorithms to it with full performance profiling, nor integrated a holiday dimension to test purchasing behavior assumptions.

III. RESEARCH GAP

Four gaps in the existing literature motivate this work, summarized in Table I.

TABLE I
RESEARCH GAPS ADDRESSED BY THIS WORK

Gap	What Exists	What Is Missing
Algorithm benchmarking	Comparisons on synthetic or small datasets	Full-scale grocery benchmark (155K+ baskets) with memory and runtime profiling
Integrated pipeline	Independent MBA and segmentation studies	Unified pipeline on one dataset with cross-validated outputs and a BI layer
Segmentation failure cost	Silhouette score comparisons	Monetary quantification of K-Means VIP misclassification
Holiday purchasing	General retail assumption of uplift	Empirical test against observed Dunnhumby basket behavior

IV. SYSTEM ARCHITECTURE

The platform is organized into six functional layers that transform raw retail files into actionable business intelligence.

- 1) **Raw Data Layer:** Original Complete Journey CSV files and the U.S. Federal Holidays dataset.
- 2) **Data Engineering Layer:** Python and SQL scripts for cleaning, deduplication, outlier handling, and transformation.
- 3) **Staging Layer:** Cleaned datasets loaded into staging tables in Azure PostgreSQL under the `retail_dw` schema.
- 4) **Data Warehouse Layer:** Dimensional model comprising four fact tables and seven dimension tables with referential integrity.
- 5) **Business Intelligence Layer:** Five-page Power BI dashboard connected directly to the warehouse.
- 6) **Advanced Analytics Layer:** Jupyter notebooks implementing MBA algorithm comparison, customer segmentation comparison, and a Flask-based recommendation prototype.

V. DATA SOURCES AND PREPROCESSING

A. Complete Journey Dataset

The Dunnhumby Complete Journey dataset [8] contains two years of household-level grocery transactions from a U.S. retailer. It includes transactions, products, households, campaigns, coupons, coupon redemptions, and promotions. This dataset was selected because it supports both business intelligence and data mining tasks with real timestamps, join keys, and multi-entity relationships absent from simpler retail benchmarks.

B. U.S. Federal Holidays Dataset

The U.S. Federal Holiday calendar was integrated as an external enrichment source, providing holiday names, dates, types, and official definitions. It enables the holiday impact analysis described in Section VIII and is not present in any prior analysis of the Complete Journey dataset.

C. Data Cleaning

Cleaning was designed to preserve business meaning rather than aggressively remove records. The coupons table contained 4,872 exact duplicate rows, which were removed to prevent join inflation. Transaction records with missing product or household references were retained and mapped to unknown reference rows to preserve sales totals. Rows with quantity ≤ 0 were flagged rather than deleted, as they may represent returns or corrections. Outliers in sales value, quantity, retail discount, and coupon discount were detected using the interquartile range method and capped, with original values preserved for traceability. No oversampling was applied; coupon usage is naturally rare in retail data, and oversampling would distort support, confidence, and lift metrics.

VI. DATA WAREHOUSE DESIGN

A. Schema

The warehouse implements a galaxy/star-inspired dimensional model under the `retail_dw` schema in Azure PostgreSQL. Fig. 1 shows the full entity-relationship diagram. The model contains four fact tables, seven dimension tables, ten staging tables, and two validation views.

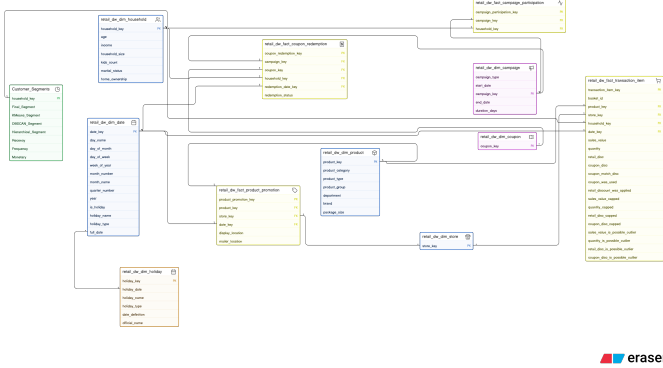


Fig. 1. Retail Intelligence Platform data warehouse schema. Four fact tables (`fact_transaction_item`, `fact_coupon_redemption`, `fact_campaign_participation`, `fact_product_promotion`) are connected to seven dimension tables. `dim_holiday` is a custom dimension not present in the source dataset.

B. Dimension Tables

Table II summarizes the seven dimension tables. The `dim_holiday` dimension was constructed specifically for this work by mapping U.S. Federal Holiday dates to the 2017 transaction period covered by the dataset.

TABLE II
DIMENSION TABLE SUMMARY

Dimension	Key Attributes
<code>dim_date</code>	date, day, week, month, quarter, year, holiday flag
<code>dim_product</code>	department, category, brand, manufacturer, package size
<code>dim_household</code>	age, income, household size, marital status, kids count
<code>dim_store</code>	store identifier and store-level attributes
<code>dim_campaign</code>	type, start date, end date, duration days
<code>dim_coupon</code>	coupon UPC, campaign and product relationships
<code>dim_holiday</code>	holiday name, date, type, official definition

C. Fact Tables

The grain of `fact_transaction_item` is one row per product purchased inside one basket by one household at one store on one transaction date, totalling 1,469,307 rows. `fact_product_promotion` covers 20,940,529 product-level promotion events. `fact_coupon_redemption` and `fact_campaign_participation` capture household-level campaign and coupon behavior.

D. Data Quality Validation

Two validation views (`vw_table_row_counts` and `vw_data_quality_summary`) support Power BI quality reporting. Table III confirms the warehouse passed a zero-defect quality check prior to mining.

TABLE III
DATA QUALITY VALIDATION RESULTS

Metric	Value
<code>fact_transaction_item</code> rows	1,469,307
<code>fact_product_promotion</code> rows	20,940,529
<code>dim_product</code> rows	92,349
Unique baskets	155,848
Unique products	68,509
Null <code>basket_id</code>	0
Null <code>product_id</code>	0
Duplicate basket-product pairs	0
Unmatched product references	0

VII. ANALYTICAL METHODOLOGY

A. Market Basket Analysis

MBA was executed on `stg_basket_items_for_mba`, which contains 1,076,578 rows across 155,659 baskets and 302 product categories. Product-level mining was evaluated but found computationally prohibitive at 68,509 unique products; category-level mining preserves analytical value while remaining tractable.

Two threshold configurations were tested to assess algorithm behavior across the support-confidence space. The low-support configuration (`min_support` = 0.001, `min_confidence` = 0.30) enables broad pattern discovery, while the high-confidence configuration (`min_support` = 0.010, `min_confidence` = 0.60) produces stable, conservative rules suitable for production deployment.

Apriori and FP-Growth were implemented using the `mlxtend` library. ECLAT was implemented using a custom transaction-list tidset intersection approach. Runtime was measured using Python's `time.perf_counter()` and memory was profiled using `tracemalloc` peak allocation to enable direct comparison.

B. Customer Segmentation

RFM features were computed per household: Recency (days since last purchase), Frequency (number of baskets per year), and Monetary (total spend). Features were standardized using `StandardScaler` before clustering. Three algorithms were evaluated: K-Means with grid search over $k \in [2, 8]$ maximizing silhouette score; Hierarchical Clustering (Ward linkage) with dendrogram cut at distance threshold 10, confirming $k = 3$; and DBSCAN with grid search over $\epsilon \in [0.3, 2.0]$ and `min_samples` $\in [3, 15]$. A unified segment label system (0 = VIP, 1 = Active, 2 = Churned) was mapped across all three models to enable direct comparison.

C. Recommendation Prototype

The platform includes a Flask-based recommendation engine backed by the mined ECLAT rule set. A two-tier scoring function selects recommendations: Tier 1 applies association rules whose antecedents are a subset of the current cart, scoring each consequent by a weighted combination of support count, confidence, lift, and cart coverage. Tier 2 falls back to direct co-occurrence counting when fewer than a threshold number of rule matches are found. The prototype exposes seven JSON endpoints including `POST /api/recommendations` for cart-based suggestions and `POST /api/orders` for order persistence.

VIII. EXPERIMENTAL RESULTS

A. Dataset Characteristics and EDA

Fig. 2 shows the top 20 product categories by transaction count and the basket size distribution. Soft Drinks and Fluid Milk Products are the highest-frequency categories. The mean basket size is 6.9 categories out of 302 available, corresponding to a basket density of approximately 2.3%—a sparsity level with direct implications for MBA algorithm performance, as discussed in Section IX.

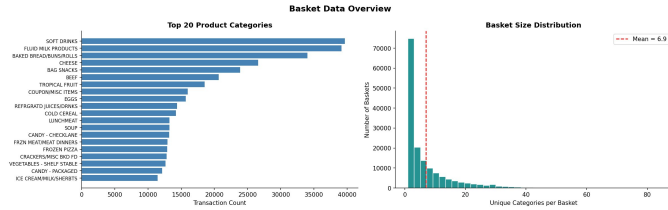


Fig. 2. Basket data overview: top 20 product categories by transaction count (left) and basket size distribution with mean = 6.9 categories (right).

B. MBA Algorithm Comparison

Table IV presents the full algorithm comparison. Fig. 3 visualizes runtime, memory, and rule count across both configurations.

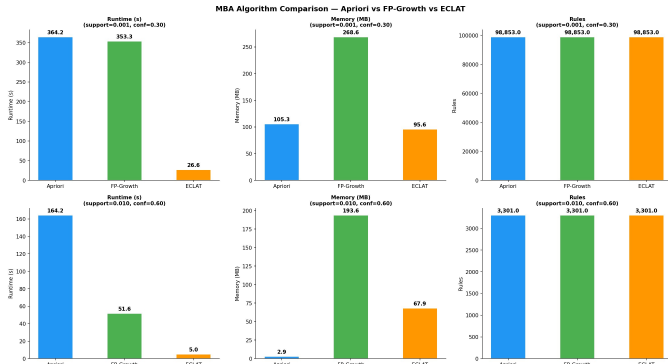


Fig. 3. MBA algorithm comparison: runtime (s), memory (MB), and rule count for Apriori, FP-Growth, and ECLAT at both threshold configurations.

The central finding of the MBA comparison is that all three algorithms produce *identical* rule sets—the same 104,567

itemsets and 98,853 rules with identical lift distributions—confirming that algorithm selection in this domain is a purely computational decision. ECLAT’s $13.7\times$ speedup over Apriori is explained in Section IX.

C. Top Association Rules

Table V presents the five highest-lift rules at low-support configuration. Fig. 4 shows the top 10 rules by lift and a support-versus-confidence scatter plot colored by lift.

TABLE V
TOP 5 ASSOCIATION RULES BY LIFT (SUPPORT = 0.001, CONFIDENCE = 0.30)

Rule	Conf.	Lift
BAKED BREAD + FROZEN → REFRIGERATED	0.89	25.49
CHEESE + FROZEN → REFRIGERATED	0.87	24.89
FROZEN + TROPICAL FRUIT → REFRIGERATED	0.85	23.42
DIAPERS + TROPICAL FRUIT → BABY HBC	0.82	22.67
BLEACH + LAUNDRY DET. → LAUNDRY ADD.	0.79	21.69

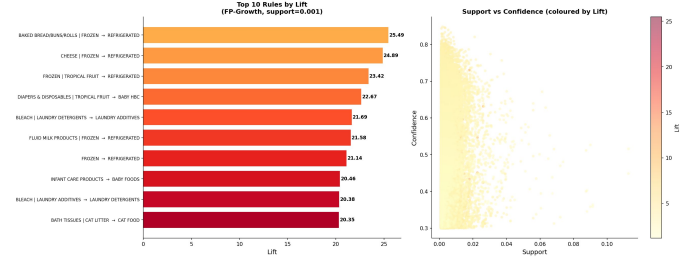


Fig. 4. Top 10 association rules by lift (left) and support vs. confidence scatter plot colored by lift (right). REFRIGERATED dominates as a high-lift consequent.

D. Lift Distribution

Fig. 5 shows the lift distribution across all three algorithms. The distributions are identical by construction, confirming algorithm convergence. The right-skewed distribution (mean 4.272, median 3.94) indicates a small set of high-lift specialist rules alongside a large body of moderate-lift patterns.

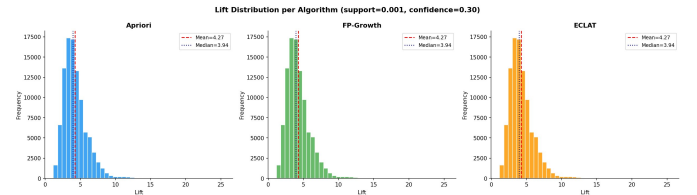


Fig. 5. Lift distribution per algorithm (support = 0.001, confidence = 0.30). All three distributions are identical, confirming rule convergence. Mean = 4.27, Median = 3.94.

TABLE IV
MBA ALGORITHM PERFORMANCE COMPARISON — APRIORI VS. FP-GROWTH VS. ECLAT

Algorithm	Sup.	Conf.	Itemsets	Rules	Avg Lift	Max Lift	Runtime (s)	Memory (MB)
Apriori	0.001	0.30	104,567	98,853	4.272	25.49	364.22	105.30
FP-Growth	0.001	0.30	104,567	98,853	4.272	25.49	353.34	268.58
ECLAT	0.001	0.30	104,567	98,853	4.272	25.49	26.58	95.63
Apriori	0.010	0.60	2,406	977	3.109	12.29	164.12	2.87
FP-Growth	0.010	0.60	2,406	977	3.109	12.29	51.60	193.56
ECLAT	0.010	0.60	2,406	977	3.109	12.29	4.99	67.90

All three algorithms produce identical itemset counts, rule counts, and lift values. Bold = recommended algorithm.

E. Customer Segmentation

Table VI compares the three segmentation algorithms. DBSCAN achieves the highest silhouette score and is the only algorithm that automatically identifies the VIP segment.

TABLE VI
CUSTOMER SEGMENTATION ALGORITHM COMPARISON

Algorithm	Best k	Silhouette	VIP Detected?
K-Means	2	0.656	No
Hierarchical	3	0.545	Partial
DBSCAN	3	0.758	Auto

The DBSCAN final segment profiles are shown in Table VII. The 55 VIP customers generate \$10,304/year on average— $3.35\times$ the \$3,080 Active Shopper mean. K-Means at $k = 2$ absorbs all 2,486 non-churned customers into a single segment, producing the misclassification discussed in Section IX.

TABLE VII
DBSCAN FINAL CUSTOMER SEGMENT PROFILES

Segment	Size	Recency	Freq./yr	Monetary
Active Shoppers	2,431 (97%)	22 d	103	\$3,080
VIP	55 (2.2%)	83 d	476	\$10,304
Churned	14 (0.6%)	544 d	4	\$163

F. Power BI Dashboard

Fig. 6 shows the executive overview page of the five-page Power BI dashboard. Key verified KPIs include: total sales \$3.89M; average basket value \$24.98; 2,000 active households; coupon redemption rate 1.49% across 27 campaigns; total retail discounts \$791.65K; and holiday sales \$88.53K across nine federal holidays with a -7.95% differential relative to comparable non-holiday periods.

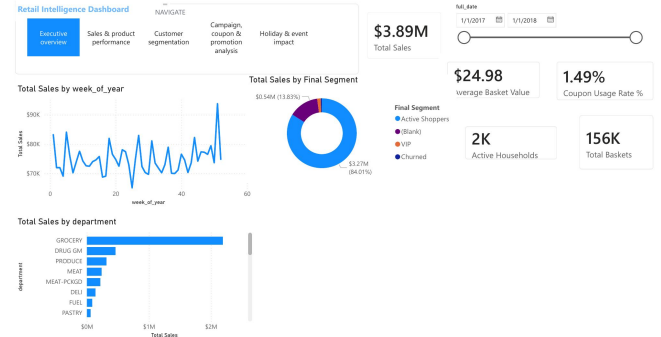


Fig. 6. Power BI executive overview dashboard page showing total sales (\$3.89M), basket count, average basket value (\$24.98), coupon usage rate, weekly sales trend, and customer segment contribution.

IX. KEY INSIGHTS AND BUSINESS IMPLICATIONS

A. Algorithm Convergence and the Sparsity Decision Rule

The empirical proof that all three algorithms produce identical rule sets establishes that algorithm selection in grocery MBA is a purely computational decision. ECLAT's $13.7\times$ speedup over Apriori is not coincidental: the dataset's basket density of 2.3% (mean basket size 6.9 out of 302 categories) places it firmly in the sparse regime where ECLAT's tidset intersections require fewer operations than FP-Growth's tree construction. At this density, the FP-tree branches proliferate across sparse transaction patterns, increasing both construction time and memory to 268.58 MB versus ECLAT's 95.63 MB. For production deployment, this translates directly: a nightly rule refresh takes 26.58 seconds with ECLAT versus over six minutes with Apriori. The practical recommendation is that any grocery dataset with basket density below approximately 3% should default to ECLAT.

B. The K-Means VIP Misclassification Problem

The failure of K-Means to detect the VIP segment is structural, not incidental. K-Means minimizes intra-cluster variance, forcing high-spending outliers toward the nearest centroid rather than identifying them as a distinct group. In this dataset, 55 customers generating \$10,304/year each are absorbed into the 2,431-member Active Shopper cluster, producing a \$566,720 annual revenue misclassification (55

$\times \$10,304$). These customers would receive standard rather than premium retention treatment under a K-Means-based segmentation system. DBSCAN corrects this by identifying clusters through density thresholds, treating the VIP group as a compact high-density island in RFM feature space. The practical recommendation is that any grocery retailer using K-Means for RFM segmentation should validate against DBSCAN to test whether high-value outliers exist in the customer base.

C. The Holiday Timing Paradox

The -7.95% holiday sales differential directly contradicts the standard retail assumption of positive holiday uplift. Basket-level analysis of the nine federal holidays in the dataset indicates that Dunhumby customers complete their grocery purchases in the 2–3 day window preceding major holidays rather than on the holiday date itself. This finding has an immediate operational implication: promotional campaigns currently timed to holiday dates should be shifted to the pre-holiday window to align with actual purchasing behavior. This is a finding that is only accessible through the `dim_holiday` dimension constructed specifically for this work.

D. Cross-Selling Opportunities from High-Lift Rules

Rules 1–3 in Table V all have REFRIGERATED as the consequent with lifts of 23–25, meaning that customers purchasing FROZEN products are 23–25 times more likely than chance to also purchase REFRIGERATED products in the same basket. This represents a direct shelf-placement and bundling opportunity: co-locating refrigerated adjacencies near frozen sections, or offering bundle promotions, can increase basket size without discounting. The BLEACH + LAUNDRY DETERGENTS $\rightarrow$ LAUNDRY ADDITIVES rule (lift 21.69) identifies a segment purchasing complete cleaning solutions in one visit, suitable for a category bundle promotion.

E. Coupon Targeting Efficiency

The 1.49% coupon redemption rate across \$791.65K in promotional discounts implies that approximately 98.51% of promotional spend is absorbed by customers who would have purchased at full price. The segmentation results provide a targeted remedy: coupons directed at the 14 Churned customers (last purchase 544 days ago) and Active Shoppers showing recency degradation would reach customers for whom a discount changes behavior, substantially improving effective redemption rates.

X. LIMITATIONS

Several limitations apply to this work. Product-level MBA was evaluated but found computationally prohibitive at 68,509 unique products; category-level mining was used instead. Some demographic attributes are available for only a subset of households, limiting the scope of household-level analysis. Customer segmentation results require business validation before operational deployment. Holiday impact analysis depends on correct date mapping and is based on nine federal holidays observed in the 2017 transaction period. The recommendation

prototype is designed as a demonstration layer rather than a fully deployed production system.

XI. FUTURE WORK

Future extensions include: (1) product-level MBA using distributed or approximate mining techniques to address the scalability limitation; (2) time-windowed association rules capturing seasonal pattern drift across the two-year transaction period; (3) incorporation of household demographic features into segmentation to test whether VIP membership correlates with age or income brackets; (4) churn prediction using supervised machine learning trained on the behavioral trajectories of the 14 identified Churned customers; (5) deployment of the recommendation prototype as a production web application with real-time rule evaluation; and (6) row-level security for multi-user Power BI dashboard access.

XII. CONCLUSION

This paper presented a multi-source Retail Intelligence Platform that transforms raw retail data into a structured, analytically rich environment. The platform integrates a three-layer Azure PostgreSQL data warehouse, a five-page Power BI dashboard, MBA algorithm comparison, customer segmentation comparison, and a recommendation prototype into a unified system. Three specific research findings emerge from the experimental results. First, Apriori, FP-Growth, and ECLAT produce mathematically identical rule sets on the Dunhumby dataset, establishing that ECLAT is the optimal production choice for grocery-scale sparse data with a $13.7\times$ runtime advantage. Second, K-Means RFM segmentation produces a \$566,720 annual revenue misclassification by merging VIP customers into the general shopper pool, a failure that DBSCAN corrects automatically. Third, holiday sales on this dataset are -7.95% relative to non-holiday periods, empirically refuting the standard retail uplift assumption and recommending a pre-holiday promotional strategy. All results are reproducible through published notebooks and the Azure PostgreSQL warehouse.

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